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COLLEGE OF COMPUTER, INFORMATION AND COMMUNICATIONS TECHNOLOGY

ICTP 318
Systems Analysis and Design

Worksheet No. 11: Work Plan		Date:
Student Names:	1. 2. 3. 4.	

Objective : Given the materials, tools and equipment, upon completion of the activity, the student must be able to:

1. Estimates project time frame.
2. Develop work breakdown structure.
3. Create staffing plan.

Background :

ESTIMATING THE PROJECT TIME FRAME

Some development methodologies have evolved in an attempt to accelerate the project through the SDLC as rapidly as possible while still producing a quality system. Regardless of whether time is a critical issue on a project or not, the project manager will have to develop a preliminary estimate of the amount of time the project will take.

Estimation is the process of assigning projected values for time and effort.

The numbers used to calculate these estimates can come from several sources. They can be provided with the methodology that is used, taken from projects with similar tasks and technologies, or provided by experienced developers. Generally speaking, the numbers should be conservative. A good practice is to keep track of the actual time and effort values during the SDLC so that numbers can be refined along the way, and the next project can benefit from real data. One of the greatest strengths of systems consulting firms is the past experience that they offer to a project; they have estimates and methodologies that have been developed and honed over time and applied to hundreds of projects.

There are two basic ways to estimate the time required to build a system. The simplest method uses the amount of time spent in the planning phase to predict the time required for the entire project. The idea is that a simple project will require little planning, and a complex project will require more planning; so using the amount of time spent in the planning phase is a reasonable way to estimate overall project time requirements.

With this approach, you take the time spent in (or estimated for) the planning phase and use industry standard percentages (or percentages from the organization's own experiences) to calculate estimates for the other SDLC phases.

DEVELOPING THE WORK PLAN

Once a project manager has a general idea of the size and approximate schedule for the project, he or she creates a **work plan**, which is a dynamic schedule that records and keeps track of all of the tasks that need to be accomplished over the course of the project. The project manager first must assemble important details about each task to be completed. Figure below shows the type of task information needed, including when it needs to be completed, the person assigned to do the work, and any deliverables that will result. The level of detail and the amount of information captured by the work plan depend on the needs of the project (and the detail usually increases as the project

progresses). Usually, the work plan is the main component of the project management software that we mentioned earlier. To create a work plan, the project manager identifies the tasks that need to be accomplished and determines how long each one will take. Then the tasks are organized within a work breakdown structure.

Identify Tasks Remember that the overall objectives for the system were recorded on the **system request**, and the project manager’s job is to identify all the tasks that will be needed to accomplish those objectives. This is a daunting task, certainly. The **methodology** that was selected by the project manager should be a valuable resource, however. The methodology that seems most appropriate for the project provides a list of steps and deliverables. A project manager can take the methodology, select the steps and deliverables that apply to the current project, and add them to the work plan. If an existing methodology is not available within the organization, methodologies can be purchased from consultants or vendors, or books like this textbook can serve as guidance. Using an existing methodology is the most popular way to create a work plan, because most organizations have a methodology that they use for projects. If a project manager prefers to begin from scratch, he or she can use a structured, top-down approach whereby high-level tasks are defined first and then broken down into subtasks. Each step is then broken down in turn and numbered in a hierarchical fashion. A list of tasks hierarchically numbered in this way is called a *work breakdown structure*, and it is the backbone of the project workplan.

Figure below shows a portion of a work breakdown structure for the design phase of an actual data warehouse development project. Each of the main tasks focuses on one of the required design deliverables. Within each task, there are subtasks listed that detail the activities required to complete the main task.

Task ID	Task Name	Duration (days)	Dependency	Status
1	Design phase	30		Open
1.1	Develop database design document	9		Open
1.1.1	Staging database design	9		Open
1.1.2	Suspense database design	9		Open
1.2	Develop rejects-handling design document	9	1.1.1, 1.1.2	Open
1.2.1	Rejects-handling engine design	9		Open
1.3	Develop OLAP design document	9	1.1.1, 1.1.2	Open
1.3.1	Universe design	9		Open
1.4	Develop OLAP design part 1	8		Open
1.4.1	High-priority reports design	8		Open
1.5	Develop application design document	9		Open
1.5.1	Group consolidation and corporate reporting (GCCR) maintenance application design	9		Open
1.6	Extract, transform, load (ETL) design document	2		Open
1.6.1	Data export utility design	2		Open
1.7	Application design document	25		Open
1.7.1	Web entry application UI design	25		Open
1.7.2	Web entry application UI design sign-off	1		Open
1.7.3	Web entry forms and database model validation	11		Open
1.8	Functional requirements document	9		Open
1.8.1	Application design	9		Open
1.8.1.1	User authentication	4		Open
1.8.1.2	Call logging	2		Open
1.8.1.3	Search	3		Open

(Thanks to Priya Padmanabhan for suggesting this example.)

The work breakdown structure can be organized in one of two ways: **by SDLC phase or by product**. For example, if a firm decided that it needed to develop a Web site, the firm could create a work breakdown structure based on the SDLC phases: planning, analysis, design, and implementation. In this case, a typical task that would take place during planning would be feasibility analysis. This task would be broken down into the different types of feasibility analysis: technical, economic, and organizational. Each of these would be further broken down into a series of subtasks. Alternatively, the firm could organize the work plan along the lines of the different products to be developed.

In the case of a Web site, for example, the products could include applets, application servers, database servers, the various sets of Web pages, a site map, and so on. Each of these products could be decomposed into the different tasks associated with the phases of the SDLC. With either approach, once the overall structure is determined, tasks are identified and included in the work breakdown structure of the work plan.

STAFFING THE PROJECT

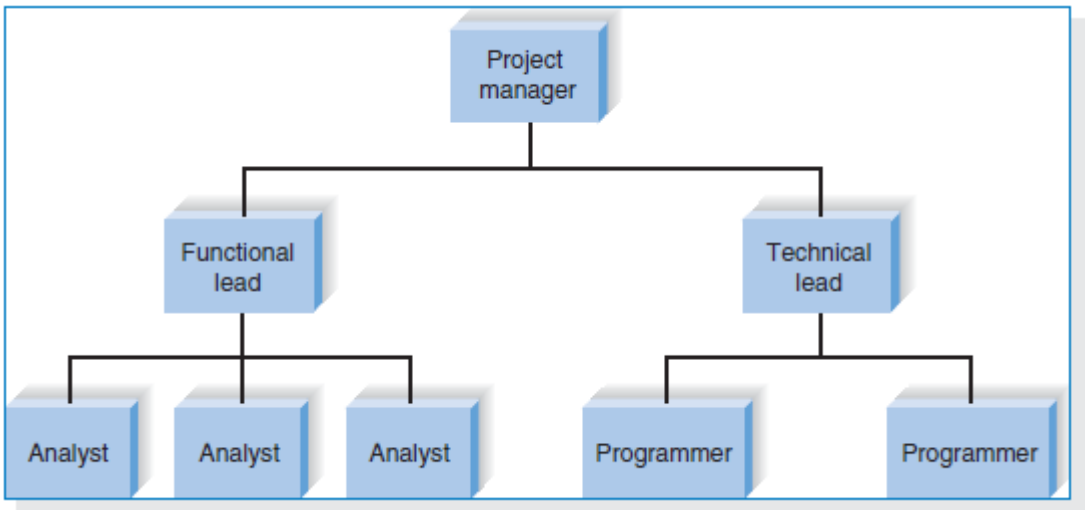
Staffing the project includes determining how many people should be assigned to the project, matching people’s skills with the needs of the project, motivating them to meet the project’s objectives, and minimizing project team conflict that will occur over time. The deliverable for this part of project management is a staffing plan, which describes the number and kinds of people who will work on the project, the overall reporting structure, and the project charter, which describes the project’s objectives and rules.

Staffing Plan

The first step to staffing is determining the average number of staff needed for the project. To calculate this figure, divide the total person-months of effort by the optimal schedule. So to complete a 40 person-month project in 10 months, a team should have an average of four full-time staff members, although this may change over time as different specialists enter and leave the team (e.g., business analysts, programmers, technical writers).

Many times, the temptation is to assign more staff to a project to shorten the project’s length, but this is not a wise move. Adding staff resources does not translate into increased productivity; staff size and productivity share a disproportionate relationship, mainly because a large number of staff members is more difficult to coordinate. The more a team grows, the more difficult it becomes to manage. Imagine how easy it is to work on a two-person project team: the team members share a single line of communication. But adding two people increases the number of communication lines to six, and greater increases lead to more dramatic gains in communication complexity.

The rule of thumb is to keep team sizes under 8 to 10 people; therefore, if more people are needed, create subteams. In this way, the project manager can keep the communication effective within small teams, which in turn communicate to a contact at a higher level in the project. After the project manager understands how many people are needed for the project, he or she creates a *staffing plan* that lists the roles that are required for the project and the proposed reporting structure for the project. Typically, a project will have one project manager who oversees the overall progress of the development effort, with the core of the team composed of the various types of analysts described in Chapter 1. A *functional lead* usually is assigned to manage a group of analysts, and a *technical lead* oversees the progress of a group of programmers and more technical staff members.



There are many structures for project teams; Figure above illustrates one possible configuration of a project team. After the roles are defined and the structure is in place,

the project manager needs to think about which people can fill each role. Often, one person fills more than one role on a project team. When you make assignments, remember that people have *technical skills* and *interpersonal skills*, and both are important on a project. Technical skills are useful for working with technical tasks (e.g., programming in Java) and in trying to understand the various roles that technology plays in the particular project (e.g., how a Web server should be configured on the basis of a projected number of hits from customers). Interpersonal skills, on the other hand, include interpersonal and communication abilities that are used when dealing with business users, senior management executives, and other members of the project team. They are particularly critical for performing the requirements-gathering activities and when addressing organizational feasibility issues. Each project will require unique technical and interpersonal skills. For example, a Web-based project may require Internet experience or Java programming knowledge, or a highly controversial project may need analysts who are particularly adept at managing political or volatile situations. Ideally, project roles are filled with people who have the right skills for the job; however, the people who fit the roles best may not be available; they may be working on other projects, or they may not exist in the company. Therefore, assigning project team members really is a combination of finding people with the appropriate skill sets and finding people who are available. When the skills of the available project team members do not match those actually required by the project, the project manager has several options to improve the situation. First, people can be pulled off other projects, and resources can be shuffled around. This is the most disruptive approach from the organization's perspective. Another approach is to use outside help—such as a consultant or contractor—to train team members and start them off on the right foot. Training classes are usually available for both technical and interpersonal instruction, if time is available. Mentoring may also be an option; a project team member can be sent to work on another similar project so that he or she can return with skills to apply to the current job.

Resources: System Analysis & Design 5th Edition by Dennis, Wixom, and Roth pages 52-61"

Accomplish the following task:

1. Develop a work plan of the proposed project. See sample estimates, work plan, and staffing plan.

The Estimates:

Industry standards suggest that a “typical” business application system spends 15% of its effort in the planning phase, 20% in the analysis phase, 35% in the design phase, and 30% in the implementation phase. This would suggest that if a project takes four months in the planning phase, then the rest of the project likely will take a total of 22.66 person-months ($4 \div .15 = 22.66$). These same industry percentages are then used to estimate the amount of time in each phase. The obvious limitation of this approach is that it can be difficult to take into account the specifics of your individual project, which may be simpler or more difficult than the “typical” project.

	Planning	Analysis	Design	Implementation
Typical industry standards for business applications	15%	20%	35%	30%
Estimates based on actual figures for first stage of SDLC	Actual: 4 person-months	Estimated: 5.33 person-months	Estimated: 9.33 person-months	Estimated: 8 person-months
SDLC = systems development life cycle.				

The Work Plan

The project work plan is the mechanism used to manage the tasks that are listed in the work breakdown structure. It is the project manager’s primary tool for managing the project. Using it, the project manager can tell whether the project is ahead of or behind schedule, how well the project was estimated, and what changes need to be made to meet the project deadline. Basically, the work plan is a table that lists all of the tasks in the work breakdown structure, along with important task information such as the people who are assigned to perform the tasks, the actual hours that the tasks took, and the variances between estimated and actual completion times. (See Figures on the succeeding pages). At a minimum, the information should include the duration of the task, the current statuses of the tasks (i.e., open, complete), and the *task dependencies*, which occur when one task cannot be performed until another task is completed. For example, It shows that task 1.2 and task 1.3 cannot begin until task 1.1 is completed. Key *milestones*, or important dates, are also identified on the work plan. Presentations to the approval committee, the start of end-user training, a company retreat, and the due date of the system prototype are the types of milestones that may be important to track.

Task ID	Task Name	Assigned To	Estimated			Actual			Dependency	Status
			Duration (days)	Start Date	Finish Date	Start Date	Finish Date	Duration variance		
1	Design Phase		31	Fri 11/16/13	Fri 12/28/13					Open
1.1	Develop database design document	Megan	9	Mon 12/3/13	Thurs 12/13/13					Open
1.1.1	Staging database design	Megan	9	Mon 12/3/13	Thurs 12/13/13					Open
1.1.2	Suspense database design	Megan	9	Mon 12/3/13	Thurs 12/13/13					Open
1.2	Develop rejectshandling design document	Megan	9	Fri 12/14/13	Wed 12/26/13				1.1.1, 1.1.2	Open
1.2.1	Rejectshandling engine design	Megan	9	Fri 12/14/13	Fri 12/26/13					Open
1.3	Develop OAP design document	Joachim	9	Fri 12/14/13	Wed 12/26/13				1.1.1, 1.1.2	Open
1.3.1	Universe design	Joachim	9	Fri 12/14/13	Wed 12/26/13					Open
1.4	Develop OAP design part 1	Kevin	8	Fri 12/7/13	Tues 12/18/13					Open
1.4.1	Highpriority reports design	Kevin	8	Fri 12/7/13	Tues 12/18/13					Open
1.5	Develop application design document	Tomas	9	Fri 12/14/13	Wed 12/26/13					Open
1.5.1	Group consolidation and corporate reporting (GCCR) maintenance application design	Tomas	9	Fri 12/14/13	Wed 12/26/13					Open

Staffing the Project (as applied to Tune Source)

Jason next turned to the task of how to staff his project. According to his earlier estimates, it appeared that about three people would be needed to deliver the system. First, he created a list of the various roles that he needed to fill. He thought he would need several analysts to work with the analysis and design of the Digital Music Download system, as well as an infrastructure analyst to manage the integration of the system with Tune Source’s existing technical environment. Jason also needed people who had good programming skills and who could be responsible for ultimately implementing the system. Kenji, Ming, and Maria are three analysts with strong technical and interpersonal skills (although Kenji is less balanced, having greater technical than interpersonal abilities), and Jason believed that they were available to bring onto this project. He wasn’t certain whether they had experience with the actual Web technology that would be used on the project, but he decided to rely on vendor training or an external consultant to build those skills later when they were needed. Because the project was so small, Jason envisioned all of the team members reporting to him because he would be serving as the project’s manager.

Jason created a staffing plan that captured this information, and he included a special incentive structure in the plan (Figure below). Rapid implementation was very important to the project’s success, so he decided to offer a day off to the team members who contributed to meeting that date. He hoped that this incentive would motivate the team to work very hard. Jason also planned to budget money for pizza and sodas for times when the team worked long hours.

Role	Description	Assigned To
Project manager	Oversees the project to ensure that it meets its objectives on time and within budget	Jason
Infrastructure analyst	Ensures that the system conforms to infrastructure standards at Tune Source; ensures that the Tune Source infrastructure can support the new system	Kenji
Systems analyst	Designs the information system—with a focus on interfaces with the CD sales system	Kenji
Systems analyst	Designs the information system—with a focus on the process models and interface design	Ming
Systems analyst	Designs the information system—with a focus on the data models and system performance	Maria
Programmer	Codes system	Ming
Programmer	Codes system	Kenji
Reporting structure: All project team members will report to Jason.		
Special incentives: If the deadline for the project is met, all team members who contributed to this goal will receive a free day off, to be taken over the holiday season.		

Before he left for the day, Jason drafted a project charter, to be fine-tuned after the team got together for its *kick-off meeting* (i.e., the first time the project team gets together). The charter listed several norms that Jason wanted to put in place from the start to eliminate any misunderstanding or problems that could come up otherwise (Figure below).

Project objective: The Digital Music Download project team will create a working Web-based system to provide digital music downloads to Tune Source’s customers as rapidly as possible.

The Digital Music Download team members will

1. Attend a staff meeting each Friday at 2 P.M. to report on the status of assigned tasks.
2. Update the work plan with actual data each Friday by 5 P.M.
3. Discuss all problems with Jason as soon as they are detected.
4. Agree to support each other when help is needed, especially for tasks that could hold back the progress of the project.
5. Post important changes to the project on the team bulletin board as they are made.